Dissecting Online Learning Mechanisms: Statistical Memory, Homeostatic Recovery, and Substrate Physics are Non-Substitutable

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Abstract

Online learning systems couple several adaptive mechanisms, but which is responsible for which capability — and whether any can be transplanted to another substrate — is rarely tested. We dissect three mechanisms of a reservoir-based online learner: a slow-trace statistical memory (M3), a homeostatic chaos regulator (M5), and the raw physics of the substrate. A falsifying transfer experiment (an echo-state network with and without the metadata under a sequential disturbance chain, 10 seeds) shows that the +32% sequential-recovery gain of the substrate system is the homeostat's, not the metadata's. The metadata does not transfer memory-capacity robustness to an ESN at any tested timescale ($\tau_m \in \{200, 500, 1000, 2000\}$; paired differences -0.68 to -0.70, 0/10 seeds positive). Its transferable value is narrower and precisely located: it denoises *state*-level corruption (gap $-0.69 \to -0.04$ to -0.01 when noise enters before the slow trace, at every tested τ_m) and accelerates boundary adaptation by a controlled-measured 1.3–2.4$\times$ on the substrate and $\sim$10 pulses on the ESN (a window-position metric artifact: 9–20$\times$). Raw memory capacity remains a substrate property (MC 10.19 vs. 6.17). Three mechanisms, three roles — none transferable to another's job. A controlled proof of concept on a Transformer with low-rank adapters confirms the boundary on a second substrate: slow-trace domain routing transfers (forgetting up to -28%, stream perplexity -1.07 at τ_m=200; 10/10 seeds) while a gating-only variant is falsified at every tested τ_m (0/10 seeds) — detection without continuous correction is insufficient.

Keywords: online learning, reservoir computing, slow features, forgetting

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kernel, disturbance robustness, mechanism disentanglement

1. Introduction

Frozen batch models — including large foundation models — cannot adapt to post-deployment reality without expensive retraining [1]; online reservoirs can [2, 4, 3], but their readouts face a timescale gap on long-horizon statistical tasks: fast reservoir states decorrelate far more quickly than the task statistics, so the readout must re-integrate after every change. Our companion work introduces a continuously active RLS readout (M1), a slow exponential trace of the reservoir states (metadata, M3), a homeostatic regulator that keeps the substrate near the edge of chaos (M5), and slow structure plasticity (M4); the integrated system tracks drift where frozen learners fail permanently (companion Paper B) [12].

These mechanisms are usually validated *together*. The question we ask here is sharper: *which mechanism does which job, and can any of them be transplanted to another substrate?* We answer with a deliberately falsifying experiment: take a generic echo-state network (ESN), give it the same metadata, run it through the same sequential disturbance chain, and measure what transfers. The answer is largely negative and, we argue, informative: the metadata is a substrate-agnostic *statistical* memory that transfers *accuracy* and *state-noise denoising*, but not *memory-capacity robustness* under readout-level disturbance; sequential recovery is the homeostat's job; raw memory capacity is the substrate's physics. Three mechanisms, three roles — none substitutable. Section 6 tests whether this decomposition survives on a structurally different substrate — a Transformer with low-rank adapters — and reports one transferable instantiation (routing) and one falsified one (gating-only).

2. Setup

Reservoir family. We compare (i) an ESN-256 with heterogeneous leak rates matched to a log-normal spectrum (CV = 0.20, spectral radius 0.9) and (ii) a Si_3N_4-style relaxation substrate whose per-unit time constants are log-normal (median $\tau_0 \approx 174\,\mu$s, CV = 0.20), driven in per-pulse contrast-coupled dynamics with coupling κ (companion Paper A) [11]. Both produce a fast state $x(t) \in \mathbb{R}^{256}$ per pulse.

Slow trace (M3). The metadata is a per-unit exponential moving average

$$m(t) = \left(1 - \frac{1}{\tau_m}\right) m(t-1) + \frac{1}{\tau_m}\, x(t), \tag{1}$$

with algorithmic timescale τ_m (pulses); the readout feature is $\varphi(t) = [x(t),\, m(t),\, 1]$.

Readout. OnlineRLS with forgetting $\lambda_f = 0.999$ (weight integration time $\tau_f = 1/(1-\lambda_f) = 1000$ pulses), predict-before-update, Tikhonov-regularized.

Tasks and metrics. *regime_switch*: three regimes with overlapping pulse-interval distributions, regime changes every $L = 1500$ pulses; single pulses are almost indistinguishable across regimes — only long-run statistics discriminate. *Mackey–Glass online prediction* under a sequential disturbance chain. Metrics: overall accuracy; switch-relative adaptation time T_{adapt} (pulses after a *known* switch to reach a 200- or 40-pulse window accuracy of 0.98/0.95); and the Jaeger held-out memory capacity (MC) probe.

Sequential disturbance chain (substrate-agnostic). Three cumulative disturbances at $t = 7\text{k}/14\text{k}/21\text{k}$ pulses: (1) timescale drift (substrate $\tau \times 1.5$; ESN leaking rate /1.5), (2) structure prune (40% of connections), (3) readout noise ($\sigma = 0.1$ on the features). Per-round MC is probed at the settled physics.

3. Theory: the slow trace is a synthesized forgetting kernel

Proposition 1 (kernel synthesis). The input-to-slow-trace channel implements a forgetting kernel with a zero at zero lag, a finite peak lag, and an exponential tail whose $1/e$ horizon is exactly τ_m — an externally controllable, substrate-independent parameter. If the fast state has kernel h_x, the slow channel kernel is the convolution of the EMA impulse response with h_x; for $h_x(u) = e^{-u/\tau_x}$,

$$h_m(\Delta t) = \frac{e^{-\Delta t/\tau_m} - e^{-\Delta t/\tau_x}}{\tau_m - \tau_x}, \qquad h_m(0) = 0, \quad \text{peak} \sim \frac{\tau_m \tau_x}{\tau_m - \tau_x} \ln \frac{\tau_m}{\tau_x}. \tag{2}$$

The material kernel of Paper A, $M(\Delta t) = \int p(\tau)\, e^{-\Delta t/\tau}\, d\tau$ (log-normal), is fixed by physics; h_m is synthesized by the algorithm, and the augmented system's effective kernel is $M \circledast h_m$ with $1/e$ horizon $\sim \tau_m + \tau_0$ (Fig. 1).

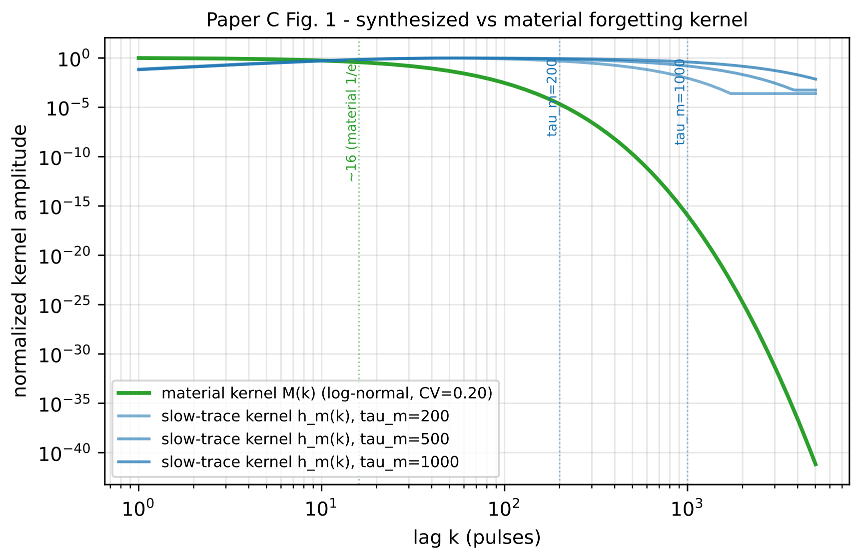

Figure 1: Synthesized vs. material forgetting kernel: the slow-trace kernel h_m (Eq. 2, $\tau_m = 200/500/1000$ pulses) has a zero at zero lag, a finite peak lag, and an exponential tail with $1/e$ horizon exactly τ_m; the material kernel M (log-normal, CV 0.20) is fixed by physics (1/e ∼16 pulses).

Proposition 2 (timescale coverage). On the statistical task the bottleneck is missing slow modes, not substrate physics. Both fast channels decorrelate at $\tau_{\text{eff}} \ll L$; the slow trace supplies the missing timescale to either substrate, so accuracy and adaptation converge to a common band. Measured (10 seeds): esn-fast 0.9955, esn-dual 0.9979, REDEM 0.9942; steady accuracy 1.000 for all (Table 1).

Proposition 3 (feature stationarity). Post-switch adaptation is bounded by the slower of feature re-centering ($\sim \tau_m + \tau_{\text{eff}}$) and weight re-integration ($\sim \tau_f$). The slow trace makes recovery feature-limited rather than weight-limited. Under a controlled switch-relative protocol, the true effect is modest but real: on the substrate the fast readout needs $T_{200} = 265$–304 pulses to re-stabilize vs. 202–211 for dual (factor 1.3–1.4 on T_{200}, 1.79–2.40 on T_{40}); on the ESN, T_{40} drops from 49.9 to 40.6 with the p90 collapsing from 76.5 to 42.0 (Table 5). The published earlier "9–20×" ratios were ratios of window positions with near-zero denominators (metric artifacts), not pulse times.

Proposition 4 (noise geometry). The EMA is a first-order low-pass

arm	overall acc	steady acc	T_{40} (pulses)
esn-fast	0.9955 ± 0.0015	1.000	49.9 ($p_{90} = 76.5$)
esn-dual (τ_m=500)	0.9979 ± 0.0001	1.000	40.6 ($p_{90} = 42.0$)
REDEM (substrate, dual)	0.9942 ± 0.0011	1.000	52.7 ($p_{90} = 67.0$)

Table 1: Equalization on regime_switch (10 seeds). Accuracy from the metadata-transfer runs; adaptation from the controlled switch-relative protocol.

filter, $|H(\omega)|^2 = 1/(1+(\omega\tau_m)^2)$. Short fast-channel transients are attenuated in the metadata channel. The transferable value of this is *denoising of state-level corruption*: when the noise enters the states before the slow trace is computed, the metadata nearly neutralizes it (variant V2 of Section 4.4). It does *not* protect against readout-level noise that hits the metadata channel equally (variant V0).

4. Experiments

4.1. Equalization on the statistical task (S10)

Table 1 reproduces the equalization result: the same slow trace ($\tau_m = 500$) closes the accuracy gap between the ESN and the substrate and collapses adaptation variance. All arms sit in the [0.994, 0.998] band with steady accuracy 1.000.

4.2. The falsifying transfer experiment (S14)

Under the sequential disturbance chain, the substrate regulated by the homeostat (M5) recovers: r_3 MC 8.47 ± 0.39 vs. 6.41 ± 0.41 for the fixed arm (+32%), with κ drifting $25.3 \rightarrow 28.5$ (Table 2; the features are fast-state only — the recovery is purely homeostatic). The ESN with the same metadata does *not* recover: paired per-seed differences $r_3\,\mathrm{MC(esn\text{-}dual)} - r_3\,\mathrm{MC(esn\text{-}fast)}$ are $-0.78/-0.76/-0.69$ at $r_1/r_2/r_3$, with 0/10 seeds positive in every round ($\sim 5\times$ the paired std). The +32% sequential recovery therefore belongs to the homeostat, not the metadata.

4.3. Pressure test across metadata timescales (S16)

Does any τ_m let the metadata close the MC gap? No: at $\tau_m \in \{200, 500, 1000, 2000\}$, the dual arm's r_3 MC is 1.04/1.05/1.06/1.06 vs. 1.74 for fast, with paired differences $-0.701/-0.693/-0.682/-0.678$ and 0/10

arm	r_0 MC	r_1 MC	r_2 MC	r_3 MC	r_3 NMSE
esn-fast	6.17	1.82	1.76	1.74	0.0277
esn-dual	6.16	1.04	1.00	1.05	**0.0251**
redem-reg	**10.19**	7.17	7.51	**8.47**	0.1063
substrate-fixed	10.19	6.47	5.97	6.41	0.1188

Table 2: Sequential disturbance chain (10 seeds). The homeostat-regulated substrate (redem-reg) reproduces the published anchor exactly (10.19/7.17/7.51/8.47); the substrate-fixed arm supplies the +32% comparison of the text.

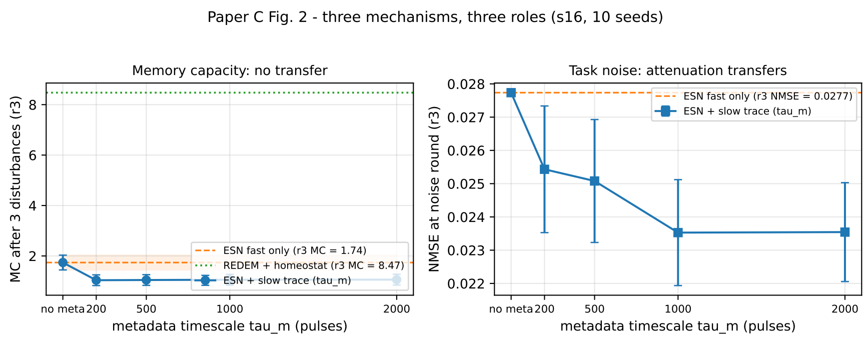

Figure 2: Post-disturbance recovery vs. metadata timescale (10 seeds). Left: r_3 MC — esn-dual stays 0.68–0.70 below esn-fast at every tested τ_m, while the homeostat-regulated substrate (redem-reg, no metadata) sits at 8.47. Right: r_3 NMSE — the readout-noise attenuation transfers at every tested τ_m.

seeds positive at every tested timescale (Table 3, Fig. 2, left panel). There is no sensitive interval; the weak upward trend (1.04→1.06) does not approach the baseline. Meanwhile the readout-noise attenuation transfers at every tested τ_m (mean-level; per-seed 10/10 at $\tau_m \geq 500$, 9/10 at τ_m=200; r_3 NMSE 0.0235–0.0254 vs. 0.0277; Fig. 2, right panel).

4.4. Probe-protocol stress test (S16b)

The falsification is robust *in sign* to the probe protocol: standardized slow features (V1) and state-level noise injection (V2) leave every (τ_m, variant) cell with at most 1/10 seeds positive (0/10 in 11 of 12 cells; the exception is the V2 cell at τ_m=200, 1/10; Table 4; the full grid data/s16b_falsification_stress_test_v2.csv extends the original $\{500, 2000\}$ pair to all four timescales). Its *magnitude* is protocol-dependent and informative: under readout-noise semantics (V0, the reference definition) the gap

τ_m	dual r_3 MC	fast r_3 MC	paired diff	n_{pos}
200	1.04	1.74	−0.701	0/10
500	1.05	1.74	−0.693	0/10
1000	1.06	1.74	−0.682	0/10
2000	1.06	1.74	−0.678	0/10

Table 3: Metadata-timescale pressure test (10 seeds): none of the tested τ_m values closes the memory-capacity gap.

variant	dual r_3 MC			paired diff
	τ_m=200	500	1000	(τ_m=200, …, 2000)
V0 readout noise	1.04	1.05	1.06	$-0.70/-0.69/-0.68/-0.68$
V1 std-slow	1.08	1.08	1.08	-0.66 (all)
V2 state noise	1.70	1.72	1.73	$-0.04/-0.02/-0.01/-0.01$

Table 4: Probe-protocol stress test (10 seeds, all four τ_m; τ_m=2000 columns omitted for the near-flat rows). Fast baseline r_3 MC 1.74; $n_{\text{pos}} \leq 1/10$ in every cell (0/10 except V2-200 at 1/10).

is -0.69; under state-noise semantics (V2) the metadata's EMA denoising nearly closes it at every tested τ_m (-0.04 to -0.01). This locates the metadata's transferable value precisely: it denoises *state*-level corruption, not readout-level noise.

4.5. Controlled adaptation (S15/S5b)

With known switch instants, the honest adaptation advantage of the metadata is a factor 1.3–2.4 on the substrate (T_{200}: fast 264.8–303.9 vs. dual/slow 202–211 pulses) and ∼10 pulses on the ESN (T_{40} 49.9→40.6; p90 76.5→42.0), with overall accuracy reproducing the original runs exactly (Table 5). A window-position metric yields inflated 9–20× ratios; those numbers are artifacts of the metric, not adaptation speedups.

5. Discussion

Three mechanisms, three roles. The evidence assembles a clean decomposition. (i) *Metadata (M3) = statistical memory*: it supplies the missing timescale on statistical tasks (Section 4.1), denoises state-level corruption (Section 4.4, V2), and attenuates readout noise on the online task

system	arm	T_{200} (pulses)	T_{40} (pulses)
substrate	fast	264.8–303.9	83.7–123.4
substrate	dual / slow	202.3–211.2	45.2–54.3
ESN	fast	210.2	49.9 ($p_{90} = 76.5$)
ESN	dual	199.2	40.6 ($p_{90} = 42.0$)

Table 5: Controlled adaptation (switch-relative pulses, 10 seeds × 5 switches): substrate ranges over both topologies, ESN values are means.

(Sections 4.2–4.3) — but it adds no episodic memory capacity (r0 MC unchanged) and cannot recover memory under readout-level disturbance at any tested τ_m (Sections 4.2–4.4). (ii) *Homeostat (M5) = disturbance recovery*: the +32% sequential recovery and the +18% $r_1 \to r_3$ gain are produced by κ-regulation alone; the metadata is not involved. (iii) *Substrate physics = raw memory*: the MC probe puts the substrate at 10.19 vs. ∼6.17 for either ESN arm; no algorithm in this work closes that lead.

Why non-transferability is informative. Each mechanism has a precisely located job, and swapping them fails in a measurable way. This delineates the boundary conditions of "metadata transfer" claims in the online-reservoir literature: transfer holds for accuracy and state-noise denoising, and does not hold for disturbance recovery or episodic memory. The same boundary reproduces on a Transformer substrate with low-rank adapters (Section 6): routing transfers, gating does not.

Related work. The slow trace is the linear slow-feature operator on reservoir states (cf. slow feature analysis [8]); the fast/slow pair is a two-timescale memory hierarchy echoing complementary learning systems [6] and synaptic consolidation [7]; the readout computes a fading-memory functional [5] whose horizon the metadata extends without touching the substrate. In the continual-learning setting [1], this work supplies causal evidence about *which* component must carry each capability.

Limitations. The slow trace's attenuation covers fast-channel (readout or state) corruption; direct corruption of the metadata memory itself is out of scope. τ_m remains a hyperparameter tuned to the task timescale. The substrate's homeostat requires a Lyapunov estimate (FTLE) that we do not provide for the ESN.

Future work. Consolidating stable, repeatedly encountered knowledge into the structural connectivity would create a hierarchy between fast tracking and slow accumulation, but introduces a consolidation-vs-overwrite trade-off beyond the present scope. Second, Section 6 instantiates the same online readout on a Transformer substrate through low-rank (LoRA-style) adapters; there the recursive-least-squares update cost is $O(F^2)$ in the readout *feature* dimension F — the model width (thousands) or an adapter rank (tens) — not $O(P^2)$ in the total parameter count (tens of billions), so the full-rank bottleneck does not arise at any scale tested here.

6. Extension to Transformer substrates: routing transfers, gating does not

6.1. Motivation and scope

The three mechanisms are formulated substrate-agnostically, but all evidence so far comes from reservoirs. We ask whether the decomposition survives on a structurally different substrate — a Transformer with low-rank adapters [9] — under the same falsification discipline (10 seeds, paired per-seed differences, sign consistency). The scope claim is deliberately narrow: the *principles* instantiate on a Transformer substrate; we make no benchmark or SOTA claims. The model is tiny by design — sized to run 90 seeded trials on CPU, not to compete with large models.

6.2. Model, task, and arms

A character-level transformer [10] with $d = 64$, two layers, two heads, context 128, and a vocabulary of 32 ($\sim 10^5$ parameters), trained online on a streaming task with known switch instants. Two biased-bigram Markov domains alternate every 3000 tokens (6 segments, 18 k tokens): domain A biases the shift-1 bigram toward symbols 0–7; domain B biases the shift-7 bigram toward symbols 24–31. Single tokens are nearly indistinguishable across domains — only bigram statistics discriminate (the analogue of the timescale gap of Section 2). Embeddings, positions, and layer norms are frozen; the output head is a trainable readout; two low-rank (rank-16) adapters on the QKV, output, and FFN projections carry the trainable plasticity.

Arms. *A1 (bare online LoRA):* both adapters update every batch at a fixed learning rate — the online-readout analogue with M1 always on. *A2 (drift gate):* a slow exponential trace of the hidden states (M3, Eq. 1) estimates the current domain; the adapter update is boosted for a window after a detected switch and suppressed within a domain — the "when to adapt" claim. *A3 (domain routing):* the same slow-trace estimate selects the active

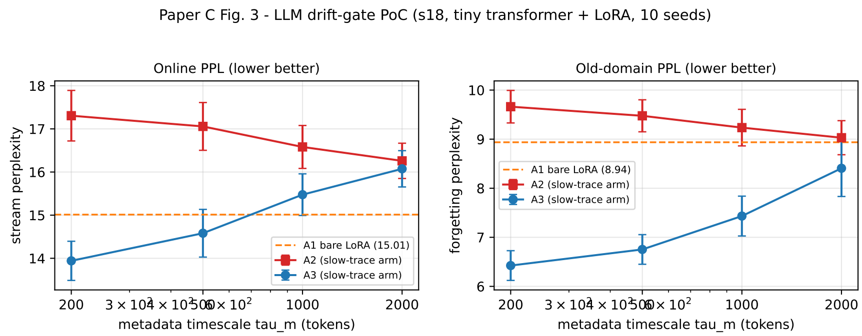

Figure 3: Transformer drift-gate proof of concept (10 seeds). Left: stream perplexity vs. metadata timescale τ_m (log scale) — A1 (bare online LoRA) as a horizontal band; A2 (gating-only) never improves; A3 (domain routing) improves at $\tau_m \in \{200, 500\}$. Right: forgetting perplexity on the previous domain — A3 improves at every tested τ_m (up to -28%); A2 does not.

adapter, and only the active one updates — the "which adapter to adapt" claim. $\tau_m \in \{200, 500, 1000, 2000\}$.

Metrics. Stream perplexity (overall); forgetting perplexity (perplexity on the previous domain after switching away — the episodic-memory probe); post-switch adaptation time in tokens (known switches, 20-token window, $1.5\times$ steady-state threshold).

6.3. Results: routing transfers, gating does not

Table 6 and Fig. 3. A3 (routing) transfers: forgetting improves at *every* tested τ_m (9–10/10 seeds, up to -2.51 ppl, -28% at τ_m=200), and stream perplexity improves when the trace is fast ($\tau_m \in \{200, 500\}$: $-1.07/-0.43$, 10/10 seeds). A2 (gating-only) is falsified: stream perplexity is worse than A1 at every tested τ_m (paired differences $+1.25$ to $+2.29$, 0/10 seeds positive) and forgetting never improves (0/10). The stream-ppl benefit of A3 is timescale-sensitive: at $\tau_m \geq 1000$ its sign flips — the same sensitive interval reported for the reservoir (Section 4.3). T_{adapt} floors at the 20-token window for every arm: the task transition is learnable within one chunk, so adaptation is too fast to resolve — reported at the floor, not as ratios (the Section 4.5 lesson). The gate detector itself is functional: it tracks the five known switches of the protocol, so the A2 failure is a failure of the update *policy*, not of the detector.

arm	τ_m	stream ppl	Δ_{ppl}	forget ppl	Δ_{forget}
A1 bare	—	15.01	—	8.94	—
A2 gate	200	17.31	+2.29 (0/10)	9.66	+0.73 (0/10)
A2 gate	500	17.06	+2.05 (0/10)	9.48	+0.54 (0/10)
A2 gate	1000	16.58	+1.57 (0/10)	9.24	+0.30 (0/10)
A2 gate	2000	16.26	+1.25 (0/10)	9.03	+0.09 (0/10)
A3 route	200	13.94	−1.07 (10/10)	6.42	−2.51 (10/10)
A3 route	500	14.58	−0.43 (10/10)	6.75	−2.18 (10/10)
A3 route	1000	15.47	+0.46 (0/10)	7.43	−1.50 (10/10)
A3 route	2000	16.07	+1.06 (0/10)	8.41	−0.53 (9/10)

Table 6: Transformer drift-gate proof of concept (10 seeds, 90 runs): stream and forgetting perplexity vs. metadata timescale τ_m; paired per-seed differences vs. A1 with $n_{\text{pos}}/10$ seeds. A3 domain routing transfers (forgetting at every tested τ_m, stream perplexity at $\tau_m \leq 500$); A2 gating-only is falsified (0/10 at every tested τ_m).

6.4. Discussion: detection without continuous correction is insufficient

The falsification of A2 is the informative result. Gating-only instantiates the metadata as a detector (M3) while suppressing the readout between detected switches: it removes M1's continuous online correction and replaces gradual switching with an abrupt boost/suppress policy. On the streaming task this loses twice — within-domain knowledge decays between switches (forgetting never improves, 0/10) and the stream pays for the suppressed updates (+1.25 to +2.29 ppl). This is the direct analogue of the reservoir finding that the metadata supplies statistical memory but not episodic memory (Section 4.2): a detector does not learn. The general design principle is that *detection without continuous online correction is insufficient*; the readout must remain active throughout the stream to maintain within-domain knowledge, and switching must be gradual (M4's "gentle wins"), not abrupt. The A3 contrast locates the transferable instantiation precisely: the slow trace's value lies in *selecting which plasticity is active*, not in *deciding when to pause learning*. The readout cost is feature-bound ($O(F^2)$ in model width or adapter rank, not $O(P^2)$ in total parameters; Section 5).

7. Conclusion

(1) A slow exponential trace of reservoir states is a synthesized forgetting kernel with a controllable horizon (Prop. 1). (2) On long-horizon statisti-

cal tasks the bottleneck is timescale coverage, not substrate physics; the same trace equalizes ESN and substrate accuracy and accelerates adaptation by a controlled-measured factor 1.3–2.4 (Props. 2–3). (3) The trace is a *statistical* memory, not an episodic one: it adds no raw memory capacity and does not transfer disturbance robustness at any tested τ_m (0/10 seeds positive; Prop. 4). (4) Three mechanisms, three roles — metadata (statistical memory and state-noise denoising), homeostat (sequential recovery), substrate physics (raw memory) — and none is substitutable for another's job. (5) The decomposition survives on a Transformer substrate with low-rank adapters: slow-trace domain routing transfers (forgetting up to −28%, stream perplexity −1.07 at τ_m=200; 10/10 seeds) while a gating-only variant is falsified at every tested τ_m (0/10 seeds) — detection without continuous correction is insufficient. (6) That boundary is a property of the *host*, not of the mechanisms: routing transfers on a Transformer substrate (point 5), and the failure of abrupt gating is attributable to its update policy — removing the continuously active readout (M1) and replacing gradual structural switching (M4) with an abrupt one. The full framework therefore does not instantiate on a frozen-feedforward host: its requirements — a continuously active online readout, a slowly integrating statistical memory, and gradually applied structural switches — are native operations of a persistent stateful substrate, not retrofits onto a static architecture. This suggests that future work should design foundation models from the ground up with these mechanisms natively embedded in a stateful host, rather than retrofitting them onto static architectures.

Declaration of competing interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Code and data availability

All simulation scripts and generated data are available at `https://github.com/huyamingc/REDEM` (private during review; released on acceptance). All reported figures are means over 10 seeds with per-seed paired analysis; data files are listed in the repository.

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